

CERTIFIED BELLMAN OPERATORS FOR COSTLY POLICY REVISION

QIAN QI

Peking University

We study the minimum implementation cost of revising an installed policy while protecting operating performance after every state and date. One randomized Markov continuation serves all restarts. Backward feasibility repair yields complete finite-state search, including a finite-horizon extension without discounting. For controlled finite models with strictly separated operating actions, disadvantage and savings coordinates contract the bilinear relaxation: a repaired root-node certificate has a quadratic small-allowance error bound at fixed primitives. An observable-covariate disintegration further gives convergent certificates for a class of controlled atomic continuous-state models without smoothing their transitions. Exact exogenous aggregation and a verified density-transfer exercise provide complementary routes. The computations retain all registered solver stops, same-object relaxation comparisons, and independent rational endpoint checks. In the original maintenance cohort, 27 lotteries beat deterministic lower bounds, while all 30 positive-cost randomized intervals remain open. The results distinguish structural rates, complete search, and actual numerical closure without changing the common-policy objective.

KEYWORDS: Dynamic programming, Policy revision, Constrained Markov control, Global optimization, Verified computation.

1. INTRODUCTION

An organization may wish to improve an installed rule without replacing every decision it makes. Implementation cost then depends both on the decisions changed today and on the states visited under the revised rule tomorrow. A feasible revision must preserve the operating objective, whereas a cost-effective revision must economize on intervention along its own induced state distribution. These two requirements create a constrained dynamic problem. Pointwise selection of inexpensive changes need not solve it.

The relevant continuation must also be specified. Two histories that meet at the same date and state use the same continuation under a Markov rule. Allowing a different continuation for each initial condition or each remembered randomization can reduce costs, but changes the economic object. Conversely, requiring a deterministic rule can increase implementation cost even when randomization is admissible. We therefore take a common randomized Markov rule as the central choice variable. Performance is protected after every date and state; implementation cost is integrated under a stated initial distribution. A performance guarantee at every restart is not a claim of cost optimality at every restart.

The main contribution is a constructive connection between these requirements and a complete global search. A policy that violates every operating constraint by at most δ can be repaired backward without adding a persistent random seed or a history variable. The repair improves operating value at every state. Its additional implementation cost is bounded explicitly by a modulus that vanishes with δ . Strict discounting provides a constant slack, while a finite-horizon allowance ladder also permits undiscounted decisions. At zero tolerance an exact restriction to operating-maximizing actions replaces the repair. No global duality assumption is used. This repair gives an error bound for rational policy lattices. For binary actions it also gives a stopping theorem for the interval Bellman branch-and-bound procedure used in the experiments.

Unlike convergence to a necessary-action relaxation, the limiting object is the original common-continuation constrained optimum.

For controlled finite models, the operating constraint itself supplies a further numerical structure. Relative to a unique operating-optimal reference, the expected probability of a non-maximizing action is bounded by the operating allowance divided by its Bellman disadvantage. Implementation savings satisfy a cone relative to operating regret. Both factors of a McCormick product therefore shrink with the allowance. A repair directed toward the reference action changes only the rare nonreference mass. These observations yield an explicit $O(\varepsilon^2)$ root-relaxation width at fixed positive operating-action separation and strict discounting. The claim concerns a certified interval for the original constrained optimum, not only approximation of an unconstrained operating value. Constants and action ties remain visible.

The continuous-state analysis is also extended beyond exogenous transitions. When an observed covariate follows an invertible autonomous evolution, it labels finite controlled problems whose optimum can be integrated exactly in principle and enclosed by verified covariate partitions. Actions may alter all regime transition probabilities. The policy uses only the current covariate and regime, so no hidden memory is introduced. Uniform finite-matrix perturbation bounds and complete fiber solves provide convergence at a fixed positive allowance. This mechanism is distinct from replacing atomic successors by smooth reset densities.

The witness-driven extension separates operating approximation error from policy-search error in one computable global bound. It supplies a positive-margin check and a constructive policy without passing the exact optimum to the repair routine. A further perturbation result converts model error into operating and cost allowances; strict randomized separations also yield break-even administration charges. These statements extend the numerical method rather than changing its economic objective.

A structural result makes a different part of the general problem computationally tractable. When state transitions are exogenous, pointwise aggregation and a compatible family of within-cell measures together preserve the optimum over unrestricted measurable common policies. These identities can hold for nonlinear atomic transitions even when total-variation reset approximation fails. The resulting all-restart regret program is linear. Twelve designed service economies, spanning three transition structures, up to 64 regimes and 64 periods, and three or five actions, produce independently verified intervals below 9×10^{-8} . An unreduced Bellman LP is an equally entitled classical comparator; we do not conceal its advantage on some small cases.

A second result transfers finite-state certificates to a continuous state space. Under a certified total-variation approximation of the transition kernel by a within-cell reset kernel, cell averaging maps an arbitrary measurable Markov policy to a finite-state policy. Lifting reverses this construction. Uniform primitive errors give an explicit operating-tolerance adjustment and a cost adjustment. The repair modulus controls the remaining difference between the relaxed and tightened finite-state optima. The theorem makes the approximation assumption operational and visible. It does not treat weak convergence of an atomic transition law as total-variation convergence, and it does not silently replace discrete shocks by continuous noise.

The numerical analysis separates three questions. First, does complete search return sound intervals for the common Markov problem, and how do those intervals compare with a direct global mathematical-programming formulation? Second, does randomization change the conclusion in the original continuous-state maintenance configurations? Third, can a nontrivial revision lower bound survive nonlinear, two-state dynamics and nonzero operating-witness error? Each experiment retains the same objective on the two sides of its interval. The inherited eight binary-action finite-state environments and their two solver implementations were fixed before those environments were generated. The additional controlled, three- and four-action

cases are separately predeclared, with a disclosed directed-arithmetic amendment; they are not described as an untouched code-frozen holdout. The original 42 configurations are explicitly a development cohort, not a prospective test.

The resulting distinctions are economically consequential. In the development cohort, 27 whole-domain randomized deployments cost strictly less than a certified lower bound for every feasible deterministic policy under the same uniform initial law. This is a statement about the value of admissible randomization, not about a neural label attached to the installed rule. The nonlinear experiment changes the zero lower bound into a positive bound; its finest horizon-32 interval is $[8.8718, 11.8492]$. That interval is informative but not tight. In the prospective finite-state comparison, both methods face the same node allowance and wall-clock limit. The stronger linear relaxation often improves the lower endpoint, but longer horizons retain substantial residual intervals. The theory supplies completeness, not a claim that complete search is inexpensive.

1.1. *Relation to the literature*

Constrained Markov decision processes provide the natural starting point (Altman, 1999). Occupation measures are particularly effective when the constraints are expectations under a designated initial law. Our formulation additionally requires every restarted problem to meet its operating guarantee using one shared continuation. A separate occupation measure for each restart would need compatibility restrictions: at a shared date and state, its action probabilities cannot depend on which restart generated the occupancy. Dropping that restriction produces a different feasible class. Our finite formulation imposes it directly through a single policy variable for each date and state. This observation does not preclude occupation-measure methods; it specifies the coupling they must preserve in this problem.

Finite approximations to constrained control are not new. Saldi (2018) proves value convergence for discounted and average-cost constrained problems, and develops approximately optimal policies and rates under additional continuity conditions. Saldi, Linder, and Yüksel (2018) provides a systematic treatment of quantized stochastic control. Our transfer result concerns finite-horizon, all-restart constraints and combines the within-cell reset identity with an explicit Markov feasibility repair. It states the total-variation condition needed to average unrestricted measurable policies rather than assuming that a convenient policy sieve is automatically dense in the constrained optimum. The theorem is not a replacement for those approximation theories; its additional object is the common-continuation restart constraint and its computable tolerance-to-cost modulus.

Primal–dual policy methods also solve constrained expected-return problems. Ding, Zhang, Basar, and Jovanovic (2020) establishes global convergence guarantees for natural policy-gradient methods in discounted constrained MDPs. Ni and Kamgarpour (2025) studies safe exploration under cumulative constraints. Those learning problems have information and feasibility requirements different from our known-model, all-restart certification problem. Safe baseline improvement (Laroche, Trichelair, and Tachet des Combes, 2019) and approximation-error analysis for fitted dynamic programming (Munos and Szepesvári, 2008, Scherrer, 2013) further distinguish model uncertainty, policy approximation, and optimization error. We claim no sample-efficiency comparison against methods given less information.

The global optimization ingredients are classical. Convex envelopes of factorable products originate in McCormick (1976); the numerical LP proposals here use a modern simplex implementation associated with Huangfu and Hall (2018). We do not attribute novelty to bilinear envelopes, best-bound search, or weak duality. The contribution is a common-policy economic formulation whose local approximation errors can be repaired into an admissible deployment,

with an explicit global stopping bound. We use the direct McCormick formulation as a serious same-object comparator, not as a deliberately weak price portfolio. A rational residual correction turns proposed LP multipliers into lower bounds even when floating-point dual feasibility is imperfect. This separation of numerical proposal and proof follows the broader program of verified computation (Rump, 2010).

Semi-infinite optimization emphasizes the difference between checking finitely many constraints and controlling them on a region (Hettich and Kortanek, 1993). Our continuous-state certificates therefore include whole-cell inequalities and isolated switching points. Information-relaxation bounds supply another route to dynamic-programming lower and upper bounds (Brown, Smith, and Sun, 2010, Brown and Smith, 2014); they are related to, but not identical with, the restart support functions used below. The support functions remain useful accelerators even though no finite price set is asserted to exhaust a global dual.

The economic motivation is adjustment rather than unconstrained operating optimization. Numerical economics has long emphasized model structure, approximation error, and state-space complexity (Judd, 1998, Brumm and Scheidegger, 2017). Empirical replacement models such as Rust (1987) discipline operating primitives with data. Our revision cost is instead the cost of departing from an installed decision rule, not a cost paid whenever the previous physical action changes. Adding physical switching costs would require including the previous action in the state. The present maintenance environments are designed, known-model economies. Their normalized operating and implementation units are not estimated monetary welfare, and their certificates are not estimates of empirical prevalence. The convergence and exact-reduction results, rather than a claimed calibration, carry the methodological argument. The density experiment now executes the reset error chain; the original endogenous-transition models remain separate evidence rather than being relabeled as instances of the exogenous theorem.

2. THE ECONOMIC PROBLEM

Let X be a standard Borel space, $A = \{1, \dots, m\}$ a finite action set, and $t = 0, \dots, T$. Rewards $r_t(x, a)$, implementation costs $k_t(x, a)$, and terminal operating payoff $g(x)$ are bounded Borel functions. The transition kernels P_t^a are known Borel probability kernels. Write $|r_t| \leq R$, $|g| \leq G$, and $0 \leq k_t \leq K$. Unless otherwise indicated, $0 < \beta < 1$ and $\varepsilon > 0$. Define

$$T_t^a v(x) = r_t(x, a) + \beta P_t^a v(x), \quad V_T = g, \quad V_t = \max_{a \in A} T_t^a V_{t+1}. \quad (1)$$

Finiteness of A makes V_t Borel and permits a Borel maximizing selector, for example the lowest indexed maximizing action.

A randomized Markov policy is a sequence of Borel kernels $p_t(\cdot | x)$ on A . Its values satisfy

$$J_T^p = g, \quad J_t^p = \sum_a p_t(a | x) T_t^a J_{t+1}^p, \quad (2)$$

$$C_T^p = 0, \quad C_t^p = \sum_a p_t(a | x) \{k_t(x, a) + \beta P_t^a C_{t+1}^p(x)\}. \quad (3)$$

The same p_t is used by all histories reaching (t, x) . Randomization is newly drawn conditional on the current date and state; a persistent hidden coin indexing different future policies is not introduced.

For an initial probability law ν , the central object is

$$K_R(\nu, \varepsilon) = \inf_{p \in \mathcal{F}_R(\varepsilon)} \int C_0^p d\nu, \quad \mathcal{F}_R(\varepsilon) = \{p : J_t^p(x) \geq V_t(x) - \varepsilon \forall t, x\}. \quad (4)$$

A deterministic policy restricts p_t to vertices of the simplex and defines K_D . An operating-optimal policy is feasible, so both classes are nonempty. On a finite state space, the product of policy simplexes is compact, J^p and C^p are continuous, and the feasible set is closed. The infimum in (4) is then attained. We use infimum, not an unproved function-space minimum, in the Borel-state formulation.

For an installed rule $a_t^0(x)$, the principal cost specification is

$$k_t(x, a) = w_t(x) \mathbf{1}\{a \neq a_t^0(x)\}, \quad w_t \geq 0. \quad (5)$$

The cost is charged for the action actually drawn. Its expectation under a lottery is not an indicator that the lottery's most probable action differs from the installed action. The unchanged installed policy has zero revision cost even when it is operating-infeasible. Consequently, dividing an optimality interval by its revision cost is undefined, not an informative relative-gap statistic. We report $(U - L)/U$ when $U > 0$, and report zero-cost exactness separately.

The uniform law in the development economy weights all normalized initial conditions equally. It is a numerical reference distribution, not an estimated stationary law. We also integrate exactly or outwardly under densities $2x$ and $2(1 - x)$. These calculations change the initial objective while leaving the all-state operating constraints and installed rule unchanged. A pointwise cost certificate would cover arbitrary initial laws, including atoms; an almost-everywhere cost enclosure need not.

3. FEASIBILITY REPAIR AND COMPLETE POLICY SEARCH

Set

$$D_C = K \sum_{s=0}^{T-1} (s+1) \beta^s, \quad (6)$$

$$D_J = \max_{1 \leq h \leq T} \left\{ 2R \sum_{s=0}^{h-1} (s+1) \beta^s + 2Gh\beta^h \right\}. \quad (7)$$

The maximum over remaining horizons in D_J matters: the terminal term $h\beta^h$ need not increase with h . Total variation between action distributions is one half of their ℓ_1 distance.

LEMMA 1—Policy perturbations: *If $\sup_{t,x} \text{TV}(p_t(\cdot | x), q_t(\cdot | x)) \leq \theta$, then*

$$\sup_{t,x} |J_t^p - J_t^q| \leq D_J \theta, \quad \sup_{t,x} |C_t^p - C_t^q| \leq D_C \theta.$$

PROOF: For costs, an action's remaining cost lies in $[0, K \sum_{s=0}^{h-1} \beta^s]$ at a state with h periods remaining. Changing its distribution changes the conditional expectation by at most the length of this interval times θ . Adding β times the next-date value difference and iterating gives $K \sum_{s=0}^{h-1} (s+1) \beta^s \theta$. For operating rewards use the interval length $2R$ for each stage and $2G$ for the terminal reward. A reward at distance s can be affected by $s+1$ action changes, and the terminal reward by h changes. Taking the maximum over h gives (7). The same conclusions follow by a finite-horizon coupling, but do not require one. Q.E.D.

THEOREM 1—Common-continuation repair: *Suppose $p \in \mathcal{F}_R(\varepsilon + \delta)$, $\delta \geq 0$. There is a Borel randomized Markov policy $\hat{p} \in \mathcal{F}_R(\varepsilon)$ with $J_t^{\hat{p}} \geq J_t^p$ for every (t, x) and*

$$\sup_{t,x} \text{TV}(\hat{p}_t, p_t) \leq \rho_\varepsilon(\delta) := \frac{\delta}{(1 - \beta)\varepsilon + \delta}, \quad \int C_0^{\hat{p}} d\nu \leq \int C_0^p d\nu + D_C \rho_\varepsilon(\delta). \quad (8)$$

In particular,

$$0 \leq K_R(\nu, \varepsilon) - K_R(\nu, \varepsilon + \delta) \leq D_C \rho_\varepsilon(\delta). \quad (9)$$

PROOF: Proceed backward, with $J_T^{\hat{p}} = g$. Given a repaired continuation, put $q_a = T_t^a J_{t+1}^{\hat{p}}$, $q_p = \sum_a p_t(a | x) q_a$, $q_* = \max_a q_a$, and $b = V_t - \varepsilon$. By the induction hypothesis, $q_p \geq J_t^p \geq V_t - \varepsilon - \delta$. Also $J_{t+1}^{\hat{p}} \geq V_{t+1} - \varepsilon$ implies $q_* \geq V_t - \beta\varepsilon$. If $q_p \geq b$, retain p_t . Otherwise choose a measurable maximizing action a_* and set

$$\hat{p}_t = (1 - \alpha)p_t + \alpha e_{a_*}, \quad \alpha = \frac{b - q_p}{q_* - q_p}.$$

Writing $d = b - q_p \in (0, \delta]$, the denominator is at least $(1 - \beta)\varepsilon + d$. Thus $\alpha \leq \rho_\varepsilon(\delta)$ and $J_t^{\hat{p}} = b \geq q_p \geq J_t^p$. All operations are Borel; the division is used only where the denominator is strictly positive. The construction uses only t and x and the already fixed next-date continuation. Lemma 1 bounds cost. Apply the construction to arbitrarily near-minimizers at tolerance $\varepsilon + \delta$ and take their infimum to obtain (9). Q.E.D.

The repair is not a convex combination of whole policies. Such a combination, if selected once and remembered, would enlarge the information state. Here the mixture is selected anew at each state using one repaired continuation. The slack $(1 - \beta)\varepsilon$ explains why the theorem assumes strict discounting and a positive tolerance. No vanishing modulus at $\beta = 1$ or $\varepsilon = 0$ is asserted by this argument.

3.1. Certified search without an exact operating solve

The exact value in the preceding statements is not an oracle that the method must possess. A verified upper witness supplies the slack needed by the repair. This matters when operating approximation, rather than policy search alone, is expensive. Let $L_t \leq V_t \leq U_t$ be bounded Borel witnesses with $U_T = g$. Define checked scalar bounds

$$\begin{aligned} e &\geq \max_{t < T} \|U_t - L_t\|_\infty, & \sigma &= (1 - \beta)\varepsilon - d, \\ d &\geq \max \left\{ 0, \sup_{t < T, x} \left[U_t(x) - \max_a T_t^a U_{t+1}(x) \right] \right\}. \end{aligned} \quad (10)$$

For the finite-state statement, let $m = |A|$, let $M \geq 1$ be an integer, and set $\theta_M = \min\{1, (m - 1)/M\}$. Lattice probabilities are multiples of $1/M$. The same primitive transition and reward operators occur in the witness check and the repair. Error in a different operator must first be included in its enclosure. A small residual is not, by itself, a certificate that $U \geq V$; the witness inequalities and terminal condition are also required.

THEOREM 2—Witness-driven repair and global error: *Suppose $\sigma > 0$ and $\delta \geq 0$. If a common randomized Markov policy satisfies $J_t^p \geq U_t - \varepsilon - \delta$ for all $t < T$, the backward construction with target $U_t - \varepsilon$ produces a common policy $\hat{p}$ such that*

$$J_t^{\hat{p}} \geq U_t - \varepsilon \geq V_t - \varepsilon, \quad J_t^{\hat{p}} \geq J_t^p, \quad \sup_{t, x} \text{TV}(\hat{p}_t, p_t) \leq \frac{\delta}{\sigma + \delta}. \quad (11)$$

Its additional initial implementation cost is at most $D_C\delta/(\sigma + \delta)$. On a finite state space, let $v_{M,L}$ be the smallest cost among rational-lattice policies satisfying $J_t^p \geq L_t - \varepsilon - D_J\theta_M$. Repair a minimizer using U , and let its cost be $U_{M,L,U}$. Then

$$\max\{0, v_{M,L} - D_C\theta_M\} \leq K_R(\nu, \varepsilon) \leq U_{M,L,U}, \quad (12)$$

$$U_{M,L,U} - \max\{0, v_{M,L} - D_C\theta_M\} \leq D_C \left\{ \theta_M + \frac{e + D_J\theta_M}{\sigma + e + D_J\theta_M} \right\}. \quad (13)$$

Thus $M \rightarrow \infty$, $e \rightarrow 0$, and $d \rightarrow 0$ give convergence to the actual constrained optimum using checked witnesses rather than an exact operating solve. All displayed errors and the stopping test are computable for rational represented finite-state primitives.

PROOF: Fix a repaired continuation and put $q_a = T_t^a J_{t+1}^{\hat{p}}$, $q_p = \sum_a p_t(a | x)q_a$, and $b = U_t - \varepsilon$. At the final decision date $J_T^{\hat{p}} = U_T$; at every earlier date the induction hypothesis gives $J_{t+1}^{\hat{p}} \geq U_{t+1} - \varepsilon$. Positivity and (10) imply

$$q_* := \max_a q_a \geq \max_a T_t^a U_{t+1} - \beta\varepsilon \geq U_t - d - \beta\varepsilon = b + \sigma.$$

The inductive improvement gives $q_p \geq J_t^p \geq b - \delta$. Where $q_p < b$, set $\alpha = (b - q_p)/(q_* - q_p)$ and mix p_t with the least-index maximizing action. For $z = b - q_p \in (0, \delta]$, one has $\alpha \leq z/(\sigma + z) \leq \delta/(\sigma + \delta)$. Otherwise leave p_t unchanged. This proves (11); the cost bound follows from Lemma 1. Finite action maximization and division on a strictly positive-denominator Borel set preserve measurability.

For the lattice statement, round any genuinely feasible policy as in Theorem 3. Its rounded value exceeds $V - \varepsilon - D_J\theta_M \geq L - \varepsilon - D_J\theta_M$, and its cost increases by at most $D_C\theta_M$. This proves the lower endpoint. A lattice minimizer is at worst $e + D_J\theta_M$ below the upper-witness target. Apply the first assertion with that value of δ to obtain the upper endpoint and (13). Nonnegative costs justify clipping the lower endpoint at zero. *Q.E.D.*

Failure of $\sigma > 0$ means that this particular witness pair has not certified the repair margin; it does not establish infeasibility of the original problem. Witness refinement and complete search have different roles in (13). Refining the witnesses alone does not close the remaining policy-search gap. Conversely, an increasingly fine policy lattice does not erase a fixed operating-witness error. The exact-value theorem is recovered by $L = U = V$ and $d = e = 0$.

3.2. A complete finite-state construction

Let X have n states. For an integer $M \geq 1$, let $\mathcal{P}_M$ be the policies whose action probabilities are multiples of $1/M$. Put $\theta_M = \min\{1, (m-1)/M\}$ and $\delta_M = D_J\theta_M$. Evaluate every candidate by the common Bellman equations and let

$$v_M = \min \left\{ \int C_0^p d\nu : p \in \mathcal{P}_M, J_t^p \geq V_t - \varepsilon - \delta_M \forall t \right\}.$$

For rational primitives these comparisons, values, and policy probabilities use rational arithmetic.

THEOREM 3—Convergence to the constrained optimum: *Repair a minimizer defining v_M and denote its feasible cost by U_M . Then*

$$L_M := \max\{0, v_M - D_C\theta_M\} \leq K_R(\nu, \varepsilon) \leq U_M, \quad U_M - L_M \leq D_C\{\theta_M + \rho_\varepsilon(\delta_M)\}. \quad (14)$$

Thus a prescribed positive absolute accuracy is reached at a finite M . At most $\binom{M+m-1}{m-1}^{nT}$ policies need be evaluated at that level. Running maxima of lower bounds and running minima of upper bounds give nested certificates without changing their validity.

PROOF: Round the first $m - 1$ probabilities of any feasible policy down to multiples of $1/M$ and assign the remainder to the last action. The total-variation error is at most $(m - 1)/M$, capped by one. Lemma 1 shows that the rounded policy belongs to the relaxed finite set and has cost at most the original cost plus $D_C\theta_M$. Hence $v_M - D_C\theta_M \leq K_R$. Theorem 1 applied to the minimizing grid policy gives $U_M \leq v_M + D_C\rho_\varepsilon(\delta_M)$. The weak-composition count gives the number of simplex points at each date and state. The displayed error tends to zero. *Q.E.D.*

The theorem controls distance to (4), not distance to a one-step relaxation. Its count is exponential in $nT(m - 1)$; it is a completeness construction, not a polynomial-time claim. The deterministic finite-state problem can instead be solved by evaluating its m^{nT} vertices with the original tolerance. The randomized repair is neither required nor valid as a deterministic-preserving operation.

3.3. The executed spatial search

For binary actions, write p_{ti} for the probability of action 1. A policy box $Q = \prod_{ti} [\ell_{ti}, u_{ti}]$ retains the same coordinates across all restarts. Backward interval Bellman evaluation gives operating upper values $\bar{J}$ and unconstrained implementation lower values $\underline{C}$ for that box. Necessary operating tests can contract its coordinate intervals: if q_0^+, q_1^+ are next-date operating upper values, an admissible p must satisfy

$$(1 - p)q_0^+ + pq_1^+ \geq V_t(i) - \varepsilon. \quad (15)$$

Intersect this half-interval with $[\ell_{ti}, u_{ti}]$. Empty intersections or inconsistent directed operating enclosures remove the box. Every removal is justified by a necessary inequality. Feasible policies are never removed because a sufficient candidate check fails.

The remaining cost lower bound is $\int \underline{C}_0 d\nu$. Repairing the midpoint of the contracted box produces a feasible upper policy. The algorithm selects a live box with the smallest lower bound and bisects a coordinate maximizing $\beta^t(u_{ti} - \ell_{ti})$. It preserves both children and every live leaf. A global lower endpoint is the minimum of all live or cost-pruned leaf lower bounds and the incumbent upper cost. It stops when their difference is no greater than the requested tolerance, or reports the remaining interval when the preassigned budget is exhausted.

PROPOSITION 1—Stopping bound for interval Bellman search: *Let $w = \max_{ti}(u_{ti} - \ell_{ti})$ for a surviving policy box. Its midpoint-repair cost minus its interval Bellman lower bound is at most*

$$H(w) = D_C\{w + \rho_\varepsilon(D_J w)\}. \quad (16)$$

Without a node or time cap, the best-bound bisection procedure reaches every prescribed positive gap η in finitely many steps. In particular, choose $w_ > 0$ with $H(w_*) \leq \eta$. A sufficient worst-case bisection depth is*

$$d \left\lceil \log_2 \frac{1}{\beta^{T-1} w_*} \right\rceil, \quad d = nT, \quad (17)$$

with the ceiling replaced by zero when its argument is at most one. The associated full binary-tree node bound is exponential in this depth. Adding valid McCormick lower bounds cannot invalidate the stopping guarantee.

PROOF: With only rectangular probability restrictions, ordinary Bellman maximization produces the maximum operating value simultaneously at every state. It is a common Markov policy, not a family of incompatible histories. A surviving contracted box has its operating upper value at least $V_t - \varepsilon$ at every date and state. Each such maximizing policy differs from the midpoint policy by total variation at most w . Lemma 1 therefore makes the midpoint $D_J w$ -approximately feasible. The same rectangular Bellman argument for minimum cost gives a lower bound no smaller than midpoint cost minus $D_C w$. Repair and (8) imply (16). A box satisfying $H(w) \leq \eta$ cannot need refinement while the global gap exceeds η : its repaired cost is already an incumbent candidate, so its lower bound is within η of the global incumbent or higher. On any still-refined path, weighted longest-side bisection makes every physical width at most w_* by the depth in (17). There are finitely many nodes up to that depth. Valid stronger lower bounds only remove or improve boxes. The supplement details the effect of the necessary contractions; they shrink the rectangles and preserve every feasible point. Q.E.D.

This is a convergence theorem for the implemented search rule, not an inference from favorable outcomes. It also explains why a 60-second run can end with a large gap: a finite worst-case bound can be prohibitively large. The numerical results report that distinction directly.

3.4. Finite-horizon repair without strict discounting

Strict discounting supplies the constant margin in Theorem 1, but it is not necessary for finite-horizon completeness. A date-dependent allowance supplies a different margin. This distinction is relevant for organizations that assess a finite service commitment without discounting its constituent dates. In this subsection only, allow $0 < \beta \leq 1$; the constants D_J, D_C retain their finite-sum definitions.

THEOREM 4—A finite-horizon slack ladder: *Let $\varepsilon > 0$, $0 < \delta < \varepsilon$, and let p satisfy $J_t^p \geq V_t - \varepsilon - \delta$ at every state and date. Choose $z \in (0, 1)$ with $\delta \leq \varepsilon z^T$ and put*

$$\gamma_t = \varepsilon z^{T-t}, \quad \tau_t = \varepsilon - \gamma_t, \quad 0 \leq t \leq T. \quad (18)$$

Backward mixing toward an action maximizing the repaired operating continuation, with current target $V_t - \tau_t$, constructs a common Borel Markov policy q satisfying

$$J_t^q \geq V_t - \tau_t \geq V_t - \varepsilon, \quad J_t^q \geq J_t^p, \quad \sup_{t,x} \text{TV}(q_t, p_t) \leq \frac{2z}{1+z}.$$

Consequently $C_0^q \leq C_0^p + D_C 2z/(1+z)$ pointwise. At $\delta = 0$ retain p ; at $\delta \geq \varepsilon$ an operating-optimal policy gives the bound D_C . With $z = (\delta/\varepsilon)^{1/T}$ the resulting modulus vanishes as $\delta \downarrow 0$, including at $\beta = 1$.

PROOF: The terminal allowance is zero. Given an already repaired continuation, write $b = V_t - \tau_t$, $q_a = T_t^a J_{t+1}^q$, $q_p = \sum_a p_t(a | x) q_a$, and $q_* = \max_a q_a$. Inductive improvement gives $q_p \geq J_t^p$, while

$$q_* - b \geq s_t := \tau_t - \beta \tau_{t+1} \geq \gamma_{t+1} - \gamma_t > 0.$$

If $q_p \geq b$, retain the current probabilities. Otherwise the mixture share $\alpha = (b - q_p)/(q_* - q_p)$ is well defined. Its deficit is at most $\delta + \gamma_t$, so

$$\alpha \leq \frac{\delta + \gamma_t}{s_t + \delta + \gamma_t} \leq \frac{\delta + z\gamma_{t+1}}{\delta + \gamma_{t+1}} \leq \frac{2z}{1+z}.$$

The last step uses $\gamma_{t+1} \geq \varepsilon z^{T-1} \geq \delta/z$ and monotonicity of $(u+z)/(u+1)$ in u . Each mixture weakly improves operating value; the first maximizing action in a fixed order is Borel. Lemma 1 yields the cost inequality. No action-independent transition assumption is used. *Q.E.D.*

Replacing the repair modulus in Theorem 3 by this modulus extends its finite-state completeness statement to $\beta = 1$. At fixed T its worst-case rate is $O(M^{-1/T})$, rather than the discounted linear rate. It is therefore an extension of the guarantee, not an assertion of efficient generic enumeration. A rational implementation need not evaluate an irrational root: a dyadic z with $\delta \leq \varepsilon z^T$ gives a completely rational construction. For $\beta < 1$, either valid repair can be used; their actual verified costs, rather than a presumed ordering of their worst-case constants, determine which candidate is retained.

The same ladder can use operating witnesses. For checked $L_t \leq V_t \leq U_t$, $U_T = g$, let d_t bound $U_t - \max_a T_t^a U_{t+1}$ from above, and choose $0 \leq \tau_t \leq \varepsilon$, $\tau_T = 0$, with

$$\sigma_t = \tau_t - \beta \tau_{t+1} - d_t > 0.$$

If $J_t^p \geq U_t - \varepsilon - \delta$, the same proof with target $U_t - \tau_t$ bounds the current mixture by $(\delta + \varepsilon - \tau_t)/(\sigma_t + \delta + \varepsilon - \tau_t)$. This is a checked, date-specific condition at $\beta = 1$, not a claim that every approximate witness has adequate slack. In a lattice bracket the deficit is $\delta = e + D_J \theta_M$. The witness-width, residual, and policy-rounding terms remain separate.

PROPOSITION 2—Zero tolerance: *At $\varepsilon = 0$, define $A_t^*(x) = \arg \max_a T_t^a V_{t+1}(x)$. The minimum implementation cost is the ordinary cost Bellman recursion restricted to $A_t^*(x)$. A deterministic Borel Markov selector attains its minimum at every state and date; randomization cannot improve it. This holds for finite T and $0 < \beta \leq 1$.*

PROOF: A feasible policy has $J_t^p = V_t$ everywhere, since no policy exceeds V_t . Thus its support is contained in A_t^* at each date, by the operating Bellman equality. Conversely every policy supported on these correspondences is operating-optimal by backward induction. Minimizing implementation cost on each nonempty finite correspondence is an unconstrained finite-action Bellman problem with Borel selectors. *Q.E.D.*

The proposition assumes the optimal-action ties are known exactly; a small numerical Bellman residual does not establish an exact tie. It addresses zero tolerance without taking a singular limit of a positive-tolerance repair bound.

4. EXACT CERTIFICATES FOR STRUCTURED ATOMIC STATE DYNAMICS

The finite search is complete but potentially expensive. A distinct question is when the common-continuation problem can be reduced exactly, without searching over probability boxes or replacing atomic successors by densities. The following conditions answer that question for exogenous state dynamics. They are restrictive and testable: neither action independence alone nor a small spatial mesh suffices.

ASSUMPTION 1—Exogenous aggregation identities: *At each date let $\{E_{ti}\}_{i=1}^{n_t}$ be a finite Borel partition and let μ_{ti} be a probability measure supported in E_{ti} . The rewards and implementation costs are constant on each cell for each action; the terminal payoff is constant on terminal cells. The transition P_t does not depend on the action. For a stochastic matrix $\bar{P}_t$,*

require both

$$P_t(x, E_{t+1,j}) = \bar{P}_{tij} \quad (x \in E_{ti}), \quad (19)$$

$$\int P_t(x, B) \mu_{ti}(dx) = \sum_j \bar{P}_{tij} \mu_{t+1,j}(B) \quad (B \text{ Borel}). \quad (20)$$

The initial law is $\nu = \sum_i \nu_i \mu_{0i}$.

Equation (19) is a pointwise identity, including cell boundaries. Equation (20) is an identity of measures, not equality only on partition cells. It allows each $P_t(x, \cdot)$ to be finitely atomic. In particular it need not be a reset kernel.

THEOREM 5—Exact common-policy aggregation: *Under Assumption 1, for finite T , $0 < \beta \leq 1$, and $\varepsilon \geq 0$, the optimum over all Borel common randomized Markov policies equals the finite-state optimum with primitives $(\bar{r}, \bar{k}, \bar{g}, \bar{P}, \nu)$. A finite optimal policy lifted constantly on cells is feasible at every original state and date and attains that continuum optimum. No regularity or cellwise-constant restriction is imposed on the original competitor policies.*

PROOF: The operating optimum is cellwise constant by (19) and backward induction. For an arbitrary Borel policy p , define $\bar{p}_t(a | i) = \int p_t(a | x) \mu_{ti}(dx)$. Action independence is essential: its operating continuation is $P_t J_{t+1}^p$, because $\sum_a p_t(a | x) = 1$. Integrating the Bellman equation over μ_{ti} and using (20) shows that the averages of J^p obey the finite Bellman equations for the one policy $\bar{p}$. The same argument applies to C^p . Every pointwise operating constraint therefore implies its finite counterpart after averaging, while the initial cost is unchanged by the mixture identity for ν . The finite infimum is no larger than the original infimum. Conversely lift any finite policy constantly on cells. Equation (19) makes its original values equal to the finite values at every point, proving feasibility and the reverse inequality. The finite optimum is attained by compactness. *Q.E.D.*

A deterministic original policy can average to a nondegenerate lottery. The theorem therefore concerns K_R , not equality between finite and continuum deterministic optima. It cannot by itself justify a deterministic lower bound for a continuum model. The deterministic separations in the original maintenance cohort continue to use their own whole-domain deterministic certificates.

4.1. An exact linear program, rather than a bilinear search

Write $\ell_{tia} = \max_b \bar{r}_{tib} - \bar{r}_{tia} \geq 0$ and let $w_0 = \nu$, $w_{t+1} = w_t \bar{P}_t$ be the exogenous state occupancies. Because the continuation is action-independent, operating regret $D_t = V_t - J_t^p$ satisfies a linear recursion. The continuum optimum in Theorem 5 is exactly the program

$$\min_{p, D} \sum_{t, i, a} \beta^t w_{ti} \bar{k}_{tia} p_{tia}, \quad (21)$$

$$D_{ti} = \sum_a \ell_{tia} p_{tia} + \beta \sum_j \bar{P}_{tij} D_{t+1,j},$$

$$\sum_a p_{tia} = 1, \quad p_{tia} \geq 0, \quad 0 \leq D_{ti} \leq \varepsilon, \quad D_T = 0.$$

For constant n, m , there are $(m + 1)nT$ variables and $2nT$ equality constraints. A transition matrix with at most s nonzeros per row gives $O(nT(m + s))$ constraint nonzeros. Rational linear programming admits polynomial bit-complexity algorithms; no such complexity claim is made for the particular simplex implementation used here. The unrestricted continuum reduction and the finite linear algebra are separate claims. The latter is ordinary linear programming, not a new global optimizer.

For clarity, the corresponding occupation-measure representation of the general, action-dependent problem has one flow y_{tia}^s for each restart $s = (u, j)$. With $z_{ti}^s = \sum_a y_{tia}^s$, it imposes initial mass one at s , the usual transition flow, and

$$y_{tia}^s = z_{ti}^s p_{tia} \quad \text{for every restart } s \text{ and } t \geq u. \quad (22)$$

Its operating constraint is $\sum_{t \geq u, i, a} \beta^{t-u} r_{tia} y_{tia}^s + \beta^{T-u} \sum_i g_i z_{Ti}^s \geq V_u(j) - \varepsilon$. A further flow initialized at ν evaluates implementation cost. Zero-occupancy states cause no division: (22) is imposed as a product equality. With controlled transitions these are genuine common-policy bilinear restrictions; independent restart occupation measures omit them. With exogenous transitions all z^s are fixed, so compatibility becomes linear. Equation (21) eliminates the redundant restart flows by retaining their regret recursion. This is the structural reason the exponential search disappears in this class.

Finite LP approximations and their convergence rates are established tools in constrained control (Dufour and Prieto-Rumeau, 2013, Saldi, 2018). The claim here is the exact preservation of the all-restart common-policy constraint under the two identities above, including atomic kernels, and its use with a repair-based certification framework. It is not priority for occupation measures, state aggregation, or linear programming.

4.2. Arithmetic certificates and economic charges

Any equality-constrained bounded LP $\min\{c'z : Az = b, l \leq z \leq u\}$ has, for every rational vector y , the lower certificate

$$L(y) = y'b + \sum_j \min\{(c - A'y)_j l_j, (c - A'y)_j u_j\}. \quad (23)$$

Subtract $y'(Az - b)$ from the objective and minimize each remaining term over its interval to prove the inequality. No exact dual feasibility or optimizer status is required. In (21), probabilities lie in $[0, 1]$ and regrets in $[0, \varepsilon]$. Rationally rounded LP probabilities are evaluated and, where needed, repaired backward; their actual cost is an upper certificate. An independent checker reconstructs both endpoints from primitives, probabilities, and multipliers. It imports neither the optimizer nor the constructor.

Exogenous state dynamics make the implementation economics particularly transparent. A nonnegative state- and action-dependent administration charge that leaves operating payoffs and transitions unchanged simply adds to $\bar{k}_{tia}$. Program (21) remains exact, rather than requiring a scalar break-even approximation. An overhead that changes rewards must also alter ℓ and the operating optimum; an overhead that changes the transition law can invalidate exogeneity and must be treated by the general common-policy program. Implementation units in the following service economies are designed adjustment units, not estimated monetary welfare.

5. REGRET-SCALED GLOBAL BOUNDS WITH ENDOGENOUS TRANSITIONS

The linear reduction above requires exogenous transitions. A different structure is available when actions change future state probabilities. Tight operating allowances limit how often

a policy can depart from an operating-maximizing action. Expressing implementation cost relative to that action then contracts both factors in the remaining bilinear products. This yields a quantitative bound for a classical McCormick relaxation, rather than only a finite-termination argument.

Throughout this section the state space is finite, $0 < \beta < 1$, and $\varepsilon > 0$. Let a_{ti}^* be the unique operating-maximizing action at (t, i) , and let H be its implementation cost. Define the operating disadvantage and the implementation advantage

$$d_{tia} = V_{ti} - r_{tia} - \beta P_t^a V_{t+1}(i), \quad (24)$$

$$g_{tia} = H_{ti} - k_{tia} - \beta P_t^a H_{t+1}(i). \quad (25)$$

Both vanish at a_{ti}^* , while $d_{tia} > 0$ for the other actions. Put $\underline{d} = \min_{t,i,a \neq a_{ti}^*} d_{tia}$. With a fixed finite model, $\underline{d} > 0$. This is a checkable separation condition, not a uniform assertion over a family in which operating actions become tied.

For a common policy p , write $D^p = V - J^p$ and $S^p = H - C^p$. Their exact recursions are

$$D_{ti}^p = \sum_a p_{tia} (d_{tia} + \beta P_t^a D_{t+1}^p(i)), \quad S_{ti}^p = \sum_a p_{tia} (g_{tia} + \beta P_t^a S_{t+1}^p(i)), \quad (26)$$

with zero terminal values. Let

$$\lambda_- = \min \left\{ 0, \min_{t,i,a \neq a_{ti}^*} \frac{g_{tia}}{d_{tia}} \right\}, \quad \lambda_+ = \max \left\{ 0, \max_{t,i,a \neq a_{ti}^*} \frac{g_{tia}}{d_{tia}} \right\}.$$

LEMMA 2—Disadvantage and savings cones: *Every feasible common policy satisfies*

$$\sum_{a \neq a_{ti}^*} d_{tia} p_{tia} \leq \varepsilon, \quad \sum_{a \neq a_{ti}^*} p_{tia} \leq \min\{1, \varepsilon/\underline{d}\}, \quad \lambda_- D_{ti}^p \leq S_{ti}^p \leq \lambda_+ D_{ti}^p. \quad (27)$$

In particular, $0 \leq D^p \leq \varepsilon$ and $\lambda_- \varepsilon \leq S^p \leq \lambda_+ \varepsilon$.

PROOF: Operating optimality gives $D^p \geq 0$. The first recursion in (26) and feasibility give the first inequality; strict action separation gives the second. Iterating the two recursions expresses D^p and S^p as discounted expected sums of d and g under the same policy and state law. The stage inequalities $\lambda_- d \leq g \leq \lambda_+ d$ also hold at the maximizing action, where both terms are zero. Taking expectations proves the cone inequalities. The last bounds use the signs of λ_- and λ_+ . Q.E.D.

Eliminate $p_{ti,a^*} = 1 - \sum_{a \neq a^*} p_{tia}$ and write $\Delta P_t^a = P_t^a - P_t^{a^*}$. The two recursions become

$$D_{ti} = \beta P_t^{a^*} D_{t+1}(i) + \sum_{a \neq a^*} p_{tia} \{d_{tia} + \beta \Delta P_t^a D_{t+1}(i)\}, \quad (28)$$

$$S_{ti} = \beta P_t^{a^*} S_{t+1}(i) + \sum_{a \neq a^*} p_{tia} \{g_{tia} + \beta \Delta P_t^a S_{t+1}(i)\}. \quad (29)$$

The objective is $\nu' H_0 - \nu' S_0$. Use the probability simplexes, the first inequality in (27), the savings cones, and ordinary nonnegative-cost bounds. Replace each remaining product $p_{tia} D_{t+1,j}$ or $p_{tia} S_{t+1,j}$ by its four McCormick inequalities over its certified rectangle. All unrestricted feasible policies remain in this linear relaxation. The construction preserves one probability variable per date, state, and action; it does not relax common continuations into restart-specific policies.

LEMMA 3—Repair of rare departures: *Suppose a policy is feasible at allowance $\varepsilon + \delta$ and its nonmaximizing action mass is at most χ at every date and state. Backward repair toward the reference action a_{ti}^* , rather than a maximizer of the repaired continuation, gives a feasible common policy with*

$$\sup_{t,i} \text{TV}(\hat{p}_{ti}, p_{ti}) \leq \chi \frac{\delta}{(1-\beta)\varepsilon + \delta}, \quad \nu' C_0^{\hat{p}} \leq \nu' C_0^p + D_C \chi \frac{\delta}{(1-\beta)\varepsilon + \delta}. \quad (30)$$

PROOF: A repaired continuation obeys $J_{t+1}^{\hat{p}} \geq V_{t+1} - \varepsilon$. Since a_{ti}^* maximizes the original operating Bellman equation, its value with that continuation is at least $V_{ti} - \beta\varepsilon$. It therefore lies strictly above the target whenever a repair is needed. The proof of Theorem 1 applies with this fixed action and bounds the mixing fraction by $\rho_\varepsilon(\delta)$. The total-variation change is exactly that fraction times $1 - p_{ti,a^*}$, at most $\chi\rho_\varepsilon(\delta)$. Operating value still improves at every backward step. Lemma 1 gives the cost bound. Q.E.D.

THEOREM 6—Quadratic small-allowance relaxation width: *Let $L_\varepsilon^{\text{LP}}$ be the exact optimum of the regret-scaled relaxation. Set $e_{D,T-1} = 0$ and, for $t < T - 1$, set*

$$e_{D,t} = \frac{\beta\varepsilon^2}{4} \max_i \sum_{a \neq a_{ti}^*} \frac{\|\Delta P_t^a(i, \cdot)\|_1}{d_{tia}}, \quad e_{S,t} = (\lambda_+ - \lambda_-)e_{D,t}, \quad e_{S,T-1} = 0.$$

Define $E_{D,T} = E_{S,T} = 0$ and $E_{D,t} = e_{D,t} + \beta E_{D,t+1}$, $E_{S,t} = e_{S,t} + \beta E_{S,t+1}$. Write $\bar{E}_D = \max_t E_{D,t}$ and $\chi_\varepsilon = \min\{1, \varepsilon/d\}$. An exact minimizing LP policy can be repaired to a feasible policy of cost $U_\varepsilon^{\text{rep}}$ such that

$$0 \leq U_\varepsilon^{\text{rep}} - L_\varepsilon^{\text{LP}} \leq E_{S,0} + D_C \chi_\varepsilon \frac{\bar{E}_D}{(1-\beta)\varepsilon + \bar{E}_D}. \quad (31)$$

For a fixed finite model with a strictly positive operating-action gap and fixed $\beta < 1$, the right side is $O(\varepsilon^2)$ as $\varepsilon \downarrow 0$.

PROOF: For any point in a McCormick rectangle, the discrepancy between its product coordinate and the actual product of its two factor coordinates is at most one quarter of the product of their interval lengths. Here a nonreference probability has length at most ε/d_{tia} . A regret coordinate has length ε , and a savings coordinate has length at most $(\lambda_+ - \lambda_-)\varepsilon$. Applying the triangle inequality to (28)–(29) gives one-step errors e_D, e_S . Recombining the reference and nonreference terms gives the stochastic kernel P^p ; its sup-norm is one. Backward comparison therefore bounds actual regret by $\varepsilon + \bar{E}_D$ and the actual policy cost by $L_\varepsilon^{\text{LP}} + E_{S,0}$. The LP's local disadvantage inequality bounds its nonreference mass by χ_ε . Lemma 3 proves (31). For fixed primitives, $E_D, E_S = O(\varepsilon^2)$ and $\chi_\varepsilon = O(\varepsilon)$; the repair ratio is $O(\varepsilon)$. Q.E.D.

This theorem does not bound the time required to solve an arbitrary LP, nor the size of a general global search. Its gain is quantitative: shrinking the operating allowance contracts both bilinear factors, and rare-departure repair prevents a first-order feasibility penalty from destroying the quadratic bound. An exact operating tie, a vanishing action gap, or a discount factor tending to one can invalidate the uniform rate. Such a case retains the ordinary finite-state certificates and complete search; it is not discarded. For a nonoptimal feasible LP proposal, its LP objective gap must be added to (31). In computation we instead report independently

checked lower and deployed-policy upper endpoints, which already include numerical proposal and rounding losses.

The use of McCormick envelopes and variable-bound tightening is classical (McCormick, 1976). The additional result is the disadvantage/savings reduction and its tolerance-dependent repair rate for the same all-restart constrained problem. The direct unscaled formulation remains a legitimate competitor. The experiment below tests the incremental contribution of these bounds without claiming dominance over every implementation of global optimization.

6. A SINGLE ERROR BUDGET FOR CONTINUOUS-STATE APPROXIMATION

A finite-state theorem alone does not justify replacing a continuous model by a grid. The following sufficient conditions permit such a transfer without restricting the original policy to be piecewise constant.

ASSUMPTION 2—Certified reset approximation: *For a finite Borel partition $(E_i)_{i=1}^{n_h}$, let μ_i be probability measures supported on E_i and write $\nu = \sum_i \nu_i \mu_i$. There are cellwise rewards $r_{h,t}$, costs $k_{h,t} \in [0, K]$, and terminal values g_h , and a kernel*

$$Q_{h,t}^a(x, dy) = \sum_j P_{h,t}^a(i, j) \mu_j(dy), \quad x \in E_i,$$

with certified uniform errors d_r, d_k, d_g and α_h satisfying

$$|r - r_h| \leq d_r, \quad |k - k_h| \leq d_k, \quad |g - g_h| \leq d_g, \quad |(P_t^a - Q_{h,t}^a)f(x)| \leq \alpha_h \|f\|_\infty$$

for every bounded Borel f . The approximation rewards obey the same bounds R, G as the original rewards, or R, G are enlarged to cover both models. All finite transition entries and error bounds are computable with verified directions.

The last inequality uses the operator convention for total variation; it is not the half- ℓ_1 convention for action probabilities above. Define $M_{J,t} = R \sum_{s=0}^{T-t-1} \beta^s + G \beta^{T-t}$ and $M_{C,t} = K \sum_{s=0}^{T-t-1} \beta^s$. Set

$$a_T = d_g, \quad a_t = d_r + \beta a_{t+1} + \beta \alpha_h M_{J,t+1}, \quad (32)$$

$$b_T = 0, \quad b_t = d_k + \beta b_{t+1} + \beta \alpha_h M_{C,t+1}, \quad (33)$$

$$a_h = \max_t a_t, \quad b_h = b_0.$$

Let $v_h(s)$ be the optimum of the finite common-Markov model at operating allowance s and initial masses (ν_i) .

THEOREM 7—End-to-end transfer: *Under Assumption 2, if $\varepsilon > 2a_h$, then*

$$v_h(\varepsilon + 2a_h) - b_h \leq K_R(\nu, \varepsilon) \leq v_h(\varepsilon - 2a_h) + b_h. \quad (34)$$

Suppose finite computation returns intervals $[L_+, U_+]$ and $[L_-, U_-]$ for these relaxed and tightened optima, of widths at most η_+ and η_- . A valid global interval is

$$[\max\{0, L_+ - b_h\}, U_- + b_h], \quad (35)$$

and its width is at most

$$\eta_+ + \eta_- + 2b_h + D_C \frac{4a_h}{(1 - \beta)(\varepsilon - 2a_h) + 4a_h}. \quad (36)$$

If all certified primitive errors tend to zero and both finite-solver widths tend to zero, this interval converges to the original measurable-policy optimum. Its measured width supplies a termination test for the actual economic objective.

PROOF: Fix any measurable Markov policy p , not necessarily cellwise constant. Backward subtraction of its original and reset-model Bellman equations gives (32)–(33). Bellman comparison gives the same operating error for optimal values. In the reset model, averaging the policy within E_i , $\bar{p}_t(a | i) = \int p_t(a | x) \mu_i(dx)$, averages its value exactly: after every transition the within-cell law is again μ_j , and stage rewards and costs are cellwise constant. This identity follows backward, including the common continuation. An original feasible policy has reset operating value at least the reset optimum minus $\varepsilon + 2a_h$ pointwise, hence also after cell averaging; its finite-model cost differs from the original initial cost by at most b_h . Taking an infimum gives the lower inequality. Conversely, lift a finite policy feasible at $\varepsilon - 2a_h$ by using its action probabilities throughout each cell. The reset value is constant in each cell; the two operating errors make the lifted original policy feasible, and its cost differs by at most b_h . This proves the upper inequality. Finally apply Theorem 1 in the finite model with target allowance $\varepsilon - 2a_h$ and violation $4a_h$, then add the two finite-computation widths and the cost errors. *Q.E.D.*

6.1. Where the transfer theorem applies

On a compact domain, a transition density with computable uniform continuity in the source variable and an L_1 -approximable density in the destination variable can furnish the reset approximation. The cell reference measures must also represent the initial law as in Assumption 2. For example, with Lebesgue-uniform cell measures and a uniform initial law, a uniformly Lipschitz transition density gives a computable $\alpha_h = O(h)$ on a shape-regular mesh. Cellwise operating and cost approximations give $d_r, d_k, d_g = O(h)$ when their moduli are Lipschitz. Aligned discontinuities in an installed rule can be retained as exact partition boundaries. In this case (36) is $O(h + \eta_+ + \eta_-)$ at fixed positive ε and $\beta < 1$, with explicitly displayed constants. The number of cells still scales as $h^{-\dim X}$; the theorem does not remove the curse of dimensionality.

The primary affine-shock model and the nonlinear two-state model below have atomic transition laws. Smearing their successor atoms into cells generally leaves the total-variation error bounded away from zero. We therefore do *not* apply Theorem 7 to those models by asserting a small mesh. Their reported intervals are direct whole-domain bounds for the unchanged atomic models. They remain valid without a convergence claim for that particular spatial representation. This distinction prevents the approximation theorem from being used as an unsupported explanation for closing an atomic-model gap.

7. CERTIFICATES THAT ACCELERATE THE GLOBAL PROBLEM

The complete search, exact structural reduction, and reset transfer provide different routes to a global certificate, each under its stated assumptions. The inherited one-sided certificates are useful accelerators and diagnostic tools, but are not substituted for that argument.

7.1. Deterministic outer restrictions and measurability

Let $L_t \leq V_t \leq U_t$ be bounded Borel operating witnesses, including the terminal date. Every all-restart feasible deterministic policy must select an action in

$$\mathcal{N}_t^{L,U}(x) = \{a : L_t(x) - T_t^a U_{t+1}(x) \leq \varepsilon\}. \quad (37)$$

Indeed, its operating continuation cannot exceed $V_{t+1} \leq U_{t+1}$. The set is nonempty because an operating-optimal action satisfies the test. Its graph is Borel, since it is a finite union of Borel sublevel sets. Backward minimization of $k_t + \beta P_t^a B_{t+1}$ over this set yields a Borel lower function; the lowest-index minimizer is a Borel selector. Thus the general-state recursion does not rely on an implicit measurable-selection assertion.

This is a necessary-action outer class, not the original constrained problem. A selected continuation can accumulate losses that violate later operating requirements. Equality between its integrated cost lower bound and a separately feasible policy's cost certifies $K_D(\nu, \varepsilon)$. Equality of the full functions certifies more. Tightening L, U alone does not force equality; complete search is what controls the remaining structural gap.

7.2. Randomized support floors and a connected local program

For $\lambda \geq 0$, a verified supersolution W_t^λ of

$$W_T^\lambda \geq \lambda g, \quad W_t^\lambda \geq \max_a \{\lambda r_t - k_t + \beta P_t^a W_{t+1}^\lambda\}$$

implies $C_t^p \geq \lambda(V_t - \varepsilon) - W_t^\lambda$ for every feasible policy, including randomized Markov policies. Define a floor F_t by the maximum of zero and finitely many such terms, using L_t in place of V_t when necessary. Given a valid next-date lower function B_{t+1} , define

$$\mathcal{D}_t(x) = \left\{ p \in \Delta(A) : \sum_a p_a T_t^a U_{t+1}(x) \geq L_t(x) - \varepsilon \right\}.$$

$$B_t(x) = \max \left\{ F_t(x), \min_{p \in \mathcal{D}_t(x)} \sum_a p_a \{k_t(x, a) + \beta P_t^a B_{t+1}(x)\} \right\}, \quad B_T = 0. \quad (38)$$

The feasible action distribution of any admissible policy belongs to the local constraint, and its continuation cost dominates B_{t+1} . Induction proves that (38) is a global randomized cost lower bound. It does not prove that the local optimizer can be combined into a feasible policy or that finite support prices close a global duality gap.

For three actions and one local operating inequality, the local polytope has only pure feasible vertices and two-action boundary mixtures. Our new primary calculation evaluates these vertices with exact rational arithmetic on directed whole-cell bounds. Unlike the earlier diagnostic with a zero global floor, it uses the nonzero F_t at every date and is paired with a feasible policy for the same uniform-initial randomized objective. It improves that lower endpoint in 23 of the 42 configurations. The prior adaptive local-LP theorem and all of its experimental records remain in the retained supplement; their target is still the local relaxation, not (4).

7.3. Whole-domain randomized deployments

A simple backward construction supplies feasible upper policies without changing the atomic model. Given a feasible next-date value J_{t+1} , write $q_0 = T_t^{a_0} J_{t+1}$, $q_* = \max_a T_t^a J_{t+1}$, and

$b = V_t - \varepsilon$. If $q_0 \geq b$, retain the installed action. Otherwise mix it with a lowest-index maximizing action with probability

$$p_t^{\text{change}}(x) = \frac{b(x) - q_0(x)}{q_*(x) - q_0(x)}. \quad (39)$$

The denominator is positive wherever this expression is used. The resulting operating value is $J_t = \max\{q_0, b\}$ and is feasible at every state. In the piecewise-affine model, J_t remains piecewise affine while the mixing probability can be a ratio of affine functions. Cost is therefore enclosed rather than falsely represented as an exact piecewise-affine function.

The verifier includes isolated switching points in every operating identity. Directed cost boxes integrate against the three absolutely continuous initial laws. All affine transition slopes are strictly positive, so a finite set of switching points and its finite-horizon preimages have zero probability under these laws. That fact justifies almost-everywhere cost enclosures; it does not justify ignoring knots in a restart operating constraint or using the same cost enclosure for a Dirac initial law.

7.4. A verified direct mathematical-programming comparator

The direct finite-state formulation minimizes $\sum_i \nu_i C_{0i}$ subject to (2)–(3), $J_{ti} \geq V_{ti} - \varepsilon$, and the common policy simplexes. The constraints are bilinear in policy probabilities and continuation values. A McCormick relaxation on each probability box provides a classical global lower program. The baseline uses the same incumbent repair, contraction rule, box selection, node allowance, initial law, and all-restart constraints as interval Bellman search. The additional work is the stronger LP bound and an LP-proposed feasible candidate after repair.

Floating-point LP status is not a certificate. For a minimization relaxation $c'z$ subject to $Az \leq b$ and $\ell \leq z \leq u$, every rational vector $\lambda \geq 0$ gives

$$\mathcal{L}(\lambda) = -\lambda'b + \sum_j \min\{(c + A'\lambda)_j \ell_j, (c + A'\lambda)_j u_j\} \leq \min c'z. \quad (40)$$

This follows by adding $\lambda'(Az - b) \leq 0$ and minimizing the resulting affine function over the box. Rounded nonnegative solver marginals are evaluated in (40) using exact rational arithmetic. A failed LP falls back to the interval Bellman lower bound; a floating infeasibility declaration never removes the policy box. A separate verifier rebuilds every relaxation and every search-tree partition from the primitive model.

8. CONTROLLED ATOMIC MODELS WITH OBSERVABLE COVARIATES

Atomic transitions need not preclude a complete continuum certificate. They do preclude treating the displacement of an atom as a small total-variation error. This section uses a different reduction: an observed covariate labels finite controlled problems, and its label remains recoverable from the current state. The regime transition is allowed to depend on the action.

ASSUMPTION 3—Observable finite fibers: *The state is $(i, z) \in \{1, \dots, n\} \times Z$, where Z is standard Borel. At date t , the next state is $(j, \Phi_t(z))$ with probability $P_t^a(i, j; z)$. Each $\Phi_t : Z \rightarrow Z$ is a Borel bijection with Borel inverse, independent of i, j, a . Primitive rewards, costs, terminal payoffs, and regime probabilities are bounded Borel functions satisfying the earlier bounds. Initially, z has law ξ and the regime has a fixed conditional law q , independent of z .*

Let F_0 be the identity and $F_t = \Phi_{t-1} \circ \dots \circ \Phi_0$. The covariate label $\zeta = F_t^{-1}(z)$ is observed at date t . For fixed ζ , evaluate the primitives at $(i, F_t(\zeta))$ to obtain a finite controlled model. Denote its common-policy constrained optimum by $v(\zeta, \varepsilon)$ under initial regime law q .

THEOREM 8—Disintegration without hidden memory: *Under Assumption 3, for $0 < \beta < 1$ and $\varepsilon > 0$,*

$$K_R(\nu, \varepsilon) = \int_Z v(\zeta, \varepsilon) \xi(d\zeta). \quad (41)$$

The infimum on the left ranges over all Borel common randomized Markov policies, not only policies that are constant on covariate cells. No exogeneity of the regime transition is required.

PROOF: A Borel policy induces the fiber policy $p_t^\zeta(a \mid i) = p_t(a \mid i, F_t(\zeta))$. The Bellman equations agree at every date and regime. An all-state feasible policy consequently has fiber cost at least $v(\zeta, \varepsilon)$ for every ζ , proving the lower inequality after integration.

For the reverse inequality, apply the rational-lattice construction of Theorem 3 separately in each fiber. At any fixed denominator, there are finitely many candidate probability arrays. Their Bellman values, relaxed-feasibility tests, lowest-index minimizers, and repaired policies are Borel in ζ . Uniform primitive bounds make the lattice and repair cost errors uniform in ζ . Thus the finite construction supplies a Borel family of exactly feasible policies whose costs approach $v(\zeta, \varepsilon)$ uniformly. It also establishes measurability of v as a limit of Borel lower endpoints. Lift this family by using $\zeta = F_t^{-1}(z)$ at the current date. Its continuation is common to every history reaching (t, i, z) ; it remembers no unobserved initial condition. Integrate its uniform approximation error and let the denominator tend to infinity. Q.E.D.

PROPOSITION 3—Computable covariate-cell enclosures: *Partition Z into finitely many Borel cells B_c and choose a representative finite fiber for each cell. Suppose primitive errors throughout cell c are bounded by $d_{r,c}, d_{k,c}, d_{g,c}$ and the finite regime matrices satisfy*

$$\sup_{t,i,a,\zeta \in B_c} \sum_j |P_t^a(i, j; F_t(\zeta)) - P_{c,t}^a(i, j)| \leq \alpha_c.$$

Compute a_c, b_c from (32)–(33), using these bounds. If $\varepsilon > 2a_c$ for every cell, and finite computations supply lower endpoints L_c^+ at allowance $\varepsilon + 2a_c$ and feasible-policy upper endpoints U_c^- at allowance $\varepsilon - 2a_c$, then

$$\sum_c \xi(B_c) \max\{0, L_c^+ - b_c\} \leq K_R(\nu, \varepsilon) \leq \sum_c \xi(B_c) (U_c^- + b_c). \quad (42)$$

If the primitive bounds and finite-solver widths tend uniformly to zero under refinement, these intervals converge to the original Borel-policy optimum at the fixed allowance ε .

PROOF: Each pair of fibers has the same finite regime coordinates. Their matrix ℓ_1 bound controls expectations of arbitrary bounded functions on these coordinates. Standard Bellman subtraction gives the operating and cost perturbation bounds, for every policy, and gives the same operating bound for optimal values. The representative model at the relaxed allowance supplies the lower bound for each true fiber; its tightened feasible policy lifts to a feasible policy in every fiber of that cell. Integrate these inequalities, using Theorem 8. The tolerance-to-cost modulus from Theorem 1, together with the finite-solver widths, proves convergence. Nonnegativity justifies truncating each lower endpoint at zero. Q.E.D.

The matrix norm in this proposition is an ℓ_1 norm on the *finite regime coordinate*, not a claim that $\delta_{\Phi_t(z)}$ and $\delta_{\Phi_t(z')}$ are close in total variation. The covariate is never smeared. A fixed observed firm type is the special case $\Phi_t(z) = z$. Nonlinear invertible covariate dynamics are also permitted. By contrast, the original maintenance transitions move the continuous condition in an action-dependent manner and do not satisfy this assumption. The fiber theorem extends the controlled atomic model class with a genuine convergence mechanism; it does not relabel the original maintenance intervals as converged.

9. NUMERICAL EVIDENCE

The experiments keep model class, information, initial distribution, and operating allowance explicit. The supplement reports every configuration; no failed or budget-stopped case is removed. The computation report records proof-object sizes, memory, wall time, the exact environment, and the boundary between independent arithmetic checks and shared-code validation.

9.1. *Controlled transitions and regret-scaled relaxation*

The controlled audit uses a condition regime $i = 0, \dots, n-1$, three or four actions, and reward $i/(n-1) - c_a$ with a small rational price perturbation. More intensive actions shift transition probability from deterioration toward improved condition and replacement. The installed rule, nonnegative revision weights, uniform initial regime law, and terminal reward are fixed by a complete rational primitive contract. Thus the transition law is genuinely endogenous, unlike the exogenous service economies below. The primitives are designed, not estimated.

Eight cases were declared before evaluation, together with solver and verifier source hashes. During the initial execution, accumulating rational expressions with unrelated denominators made certificate arithmetic unnecessarily expensive; an integer-string limit also interrupted a run. The replacement accumulates outward dyadic enclosures of individual summands. Original source, results, the amendment, and both remote hash registrations remain in the computation record. No case, operating allowance, action set, transition generator, or LP proposal rule was changed. This is a predeclared-case experiment with a disclosed arithmetic amendment, not an untouched code-frozen holdout.

Both methods use the reference-eliminated Bellman formulation, identical operating-regret bounds and probability caps, the same LP engine and tolerances, and the same exact candidate repair. Method U uses ordinary implementation-cost bounds. Method S additionally uses the savings cones and contracted savings bounds. This is a same-object root-relaxation ablation; neither method branches in this experiment. The 30-second cap applies to the LP call, not to model construction or proof checking. All those costs are recorded separately. A stopped LP contributes a sound fallback bound and the operating-optimal feasible reference, not an unverified primal or dual estimate.

The savings bounds materially reduce the certified interval in 5 of the eight pairs (Table I); a reduction below 10^{-8} in absolute width is not counted. The eight-regime, 32-period, four-action case at allowance 10^{-3} gives $[16.213382, 16.258594]$, a relative width of 0.279%. Other cases receive little benefit, and 4 LP calls reach their cap. In particular, a stronger mathematical relaxation need not give a stronger *computed* certificate within a fixed time limit. Intersecting any independently valid intervals is permitted, but the table retains the individual methods rather than hiding an unfavorable run behind a portfolio.

At allowance 10^{-4} the scaled positive-cost interval has width at most 0.000008657, compared with 0.009521686 for the unscaled relaxation. This calculation illustrates why tight allowances can help a separated-action formulation even though they worsen the generic repair denominator.

TABLE I
ALL PREDECLARED CONTROLLED CASES: ROOT-RELAXATION CERTIFICATES

Case	n	T	m	Method	Lower	Upper	Gap/upper (%)	LP status
00	3	4	3	U	2.628082	2.718904	3.341	Solved
00	3	4	3	S	2.652991	2.712135	2.181	Solved
01	8	8	3	U	4.948069	5.284000	6.358	Solved
01	8	8	3	S	5.167607	5.235970	1.306	Solved
02	16	12	3	U	4.859588	5.388508	9.816	Solved
02	16	12	3	S	4.859588	5.388508	9.816	Solved
03	16	24	4	U	17.921643	19.730330	9.168	Solved
03	16	24	4	S	18.516903	19.416854	4.635	Solved
04	32	16	3	U	5.364739	5.495882	2.387	Solved
04	32	16	3	S	5.384431	5.485796	1.848	Solved
05	8	32	4	U	15.974376	16.278277	1.867	Solved
05	8	32	4	S	16.213382	16.258594	0.279	Solved
06	16	32	4	U	0.000000	28.613836	100.000	Cap
06	16	32	4	S	0.000000	28.613836	100.000	Cap
07	32	32	3	U	0.000000	15.797290	100.000	Cap
07	32	32	3	S	0.000000	15.797290	100.000	Cap

U: unscaled. S: regret-scaled. Endpoints are outward decimal displays of independently checked rational endpoints. “Cap” retains the complete sound fallback interval. These are root bounds for the same constrained common-policy optimum, not floating-point LP objective intervals. The supplement reports all allowances, dimensions, timings, memory, proof sizes, and worst-case constants.

TABLE II
THREE ORDERS OF ALLOWANCE REFINEMENT IN ONE FIXED EIGHT-PERIOD MODEL

ε	U width	S width	S width/ ε^2	S total (s)	S check (s)
0.1000	2.181082557	2.181330051	218.13	0.049	0.035
0.0100	0.194637170	0.174925073	1749.3	0.045	0.035
0.0010	0.030918109	0.004397164	4397.2	0.044	0.035
0.0001	0.009521686	0.000008657	865.63	0.043	0.035

The model has three regimes and three actions. U and S are defined in Table I. Total time includes primitive construction, LP proposal, and constructor arithmetic; the independent check is additional. Widths are displayed upward. The ratio is descriptive: the theorem is a fixed-model upper rate, not an assertion that every pair of observed widths has slope two.

It does not imply that the action-separation constants are uniformly favorable. The supplement reports $\underline{d}$, the savings-cone slopes, and each term of (31); these sufficient bounds can be much larger than the measured interval. None of the eight larger-state cases is relabeled as having met an absolute target merely because its relative gap is small.

9.2. An executed controlled atomic continuum certificate

Consider a continuum of observable establishment sizes $z \in [0, 1]$, fixed over the decision horizon. Condition evolves according to the four-regime, three-action controlled maintenance law specified in the replication contract; the horizon is eight and $\beta = 0.95$. Operating rewards and the terminal reward are multiplied by $s(z) = 1 + z$, while revision costs are multiplied by $w(z) = 1 + z/2$. Size and the initially uniform regime are independent, and size is uniform. The

TABLE III
CONTROLLED ATOMIC CONTINUUM: PAIRED PARTITION BOUNDS

Cells	Lower	Upper	Gap/upper (%)	Finite widths	Partition remainder
2	9.926685	10.269496	3.339	0.056918	0.285893
4	10.015768	10.223818	2.035	0.066491	0.141558
8	10.055476	10.198274	1.401	0.072194	0.070603
16	10.074215	10.184476	1.083	0.074982	0.035279

All 17 finite endpoint problems and all four weighted sums are independently checked. “Finite widths” is the sum of the upper-endpoint fiber widths with exact cell weights. The remainder is the displayed total interval width minus that quantity; it includes value change and differences between adjacent lower endpoints, not an independently estimated quadrature error.

operating allowance is 0.01 for every size, condition, and restart. Every transition is atomic in the continuous size coordinate. Actions nonetheless change the condition transition probabilities.

Writing v_0 for the base finite-model optimum, Theorem 8 gives

$$K_R = \int_0^1 (1 + z/2) v_0 \left(\frac{0.01}{1 + z} \right) dz. \quad (43)$$

This is an optimization identity, not an assumed restriction to a parametric policy. On a cell $[a, b]$, monotonicity of v_0 in its allowance supplies a lower endpoint from the finite solve at $0.01/(1 + a)$ and a feasible upper policy from the tighter solve at $0.01/(1 + b)$. Multiply these endpoints by the exact positive weight $(b - a) + (b^2 - a^2)/4$ and sum. The upper policy uses the observed size cell at every date and retains common continuations.

The 16-cell interval is $[10.074215, 10.184476]$, with relative width 1.083%. It is a paired global certificate for an uncountable-state controlled atomic model, not a density approximation or a collection of pointwise policy tests. Partition refinement reduces the displayed interval, but the finite root widths remain material. Arbitrarily refining size cells alone does not force this execution to reach zero width: the finite fibers must also be solved more accurately, as required by Proposition 3. The original action-dependent continuous-condition maintenance models are different models and retain their existing status.

9.3. Closed atomic continuous-state service economies

Consider an organization selecting one of m service intensities $a/(m - 1)$ in demand regime i . Normalized demand is $q_i = (2i + 1)/(2n)$, operating reward is $-(q_i - a/(m - 1))^2$, and terminal payoff is zero. The installed rule is one intensity below the lowest-index reward-maximizing action, truncated at zero. Keeping that rule has zero implementation cost; a realized departure costs $1 + q_i + |a - a_i^0|/(m - 1)$. Demand is exogenous to service intensity. Initial regime masses are $2(i + 1)/(n(n + 1))$. Banded, cyclic, and dense transition matrices are specified exactly in the supplement. These are designed known-model economies, not calibrated staffing data.

The state is not merely a finite regime. It is $(i, u, v) \in \{0, \dots, n - 1\} \times [0, 1]^2$. Conditional on a regime transition $i \rightarrow j$, the continuous coordinates move deterministically according to

$$u' = (u + v^2) \bmod 1, \quad v' = (v + (u')^2) \bmod 1. \quad (44)$$

Each shear is a translation on one circle conditional on the other coordinate and hence preserves uniform measure; their composition does too. Taking uniform measures on the two-dimensional cells verifies both aggregation identities exactly. Every transition law is finitely atomic, including

TABLE IV
EXACT AGGREGATION OF ATOMIC CONTINUUM MODELS: ALL REGISTERED CASES

n	T	m	Law	β	ε	Lower	Upper	Gap
8	8	3	B	0.95	0.01	13.120621355	13.120621379	2.291×10^{-8}
16	16	3	B	0.95	0.01	22.366537651	22.366537682	2.938×10^{-8}
32	32	3	B	0.95	0.01	32.576896364	32.576896401	3.589×10^{-8}
64	64	3	B	0.95	0.01	39.128135567	39.128135608	4.013×10^{-8}
8	8	3	C	0.99	0.05	12.174612806	12.174612832	2.535×10^{-8}
16	16	3	C	0.99	0.05	23.440340924	23.440340965	4.034×10^{-8}
32	32	3	C	0.99	0.05	43.707336818	43.707336878	5.966×10^{-8}
16	16	3	D	0.95	0.01	18.290633115	18.290633156	3.952×10^{-8}
8	8	3	B	1	0.01	15.605690050	15.605690075	2.431×10^{-8}
16	16	3	B	1	0.01	32.013733752	32.013733787	3.352×10^{-8}
32	32	3	C	1	0.05	50.441701001	50.441701067	6.497×10^{-8}
16	16	5	B	0.95	0.01	20.009468627	20.009468717	8.820×10^{-8}

Endpoints are rounded outward to nine decimal places; the last column bounds the unrounded arithmetic gap. B, C, and D denote banded, cyclic, and dense regime laws. Both continuous coordinates remain atomic under (44). Independent checks concern finite arithmetic; measure preservation and the aggregation theorem are analytic proofs.

at discontinuities of the modulo representation. The theorem applies to arbitrary policies that can observe and use both continuous coordinates; it does not assume them away by restricting policies in advance. Payoffs depend on the regime, so this is a structured continuum application rather than evidence of general high-dimensional numerical performance.

The twelve cases were registered on the new branch before execution, including the 60-second LP limit and absolute certificate target 10^{-4} . Source code was not frozen by that protocol commit; this experiment is consequently distinguished from the earlier fully code-frozen prospective suite. No case was removed or replaced. Table IV reports all twelve. Every pair of endpoints is independently reconstructed in rational arithmetic. All costs are positive, and every interval meets the registered target.

The largest program has 16,384 variables and 8,192 equalities, corresponding to 4,096 state–date policy simplexes. The supplement separates model assembly, LP proposal, exact certificate construction, independent checking, and process memory. It also reports the same-object unreduced Bellman LP with ordinary presolve. Neither formulation branches on probabilities. Structure elimination is faster on several larger rows, whereas the unreduced LP is faster on some small rows. These observations establish tractable closure for this class; they do not demonstrate superiority over the best classical LP formulation or over the earlier McCormick code on its different, action-dependent problems.

9.4. An executed nonzero total-variation transfer budget

For an additional continuous-state model keep the service economy but replace (44) by the within-cell density

$$P(i, u; \{j\} \times dv) = \bar{P}_{ij}[1 + \zeta f(u)f(v)] dv, \quad f(u) = 2u_1 - 1, \quad 0 \leq \zeta \leq 1.$$

Here u, v each denote a point of $[0, 1)^2$. This density is nonnegative and integrates to one. Integrating over a uniform source cell gives a uniform destination cell, so the exact aggregation theorem supplies an independent structural reference optimum. The reset approximation sets $\zeta = 0$. Its operator error, in the convention of Theorem 7, is $\alpha_h = \zeta/2$; its probability total

TABLE V
MEASURED CONTINUOUS TRANSFER AND ITS EXPLICIT ERROR TERMS

ζ	Operator α_h	a_h	$2b_h$	Transferred width	Status
0.1	0.05	0.955941	5.846110	—	condition fails
0.001	0.0005	0.009559	0.058461	0.929921	paired
1e-06	5e-07	0.000010	0.000058	0.000934	paired

Successful rows use independently checked finite lower and feasible upper policies at allowances $\varepsilon + 2a_h$ and $\varepsilon - 2a_h$. Failure of the sufficient reset condition is not economic infeasibility: the exact structural reference remains available.

variation is $\zeta/4$. Reward, implementation, and terminal approximation errors are all zero. The initial law uses exactly the same uniform reference measures.

For $n = T = 8$, $m = 3$, $\beta = 0.95$, and $\varepsilon = 0.05$, we construct every primitive and propagated error and solve both adjusted finite problems whenever $\varepsilon > 2a_h$. Table V retains the failed tightening condition as well as the successful transfers. The supplement gives the two finite-solver widths, $2b_h$, and the repair-cost bound separately. In particular a small measured transfer interval must not be confused with a small worst-case repair bound.

This exercise instantiates the reset theorem rather than substituting a mesh trend for its assumptions. It also exhibits its conditioning: the largest density perturbation defeats the prescribed tightening, and the next one gives a wide interval. The exact aggregation result does not incur this transfer error. Neither result is applied to the original action-dependent maintenance kernels, which do not satisfy the new exogeneity assumption.

9.5. Prospectively fixed common-Markov problems

Before generating eight new environments, we committed their seeds, joint parameter changes, two solver hashes, node cap 4,096, wall-clock cap 60 seconds per solver and case, and absolute stopping allowance 10^{-3} . The freeze is commit 13fd3ee; it follows development seed 390001 and precedes all eight evaluated seeds. Both solvers see the same known model. The two-state models represent low and high condition; four-state models use a pair of condition indicators with interaction terms in their transition probabilities. Horizons are 3, 4, 8, and 12; discount factors are 0.8 or 0.95; allowances are 0.01 or 0.05; operating prices, transition persistence, intervention effects, and initial masses vary jointly. The exact formulas and resulting rational matrices are in the supplement and proof objects.

Table VI contains all eight pairs, including unfinished searches. A positive absolute-gap certificate is not necessarily a small relative-gap certificate when the optimum is small. We therefore report both. The linear relaxation often improves the lower endpoint substantially, but costs more per node. The four-state longer-horizon cases expose the combinatorial burden: the remaining interval is not hidden by a success count. The comparison does not establish that interval Bellman search dominates direct global programming. Instead it identifies the value and cost of the stronger relaxation under matched information and caps.

9.6. The unchanged continuous-state development cohort

The original maintenance model has $x \in [0, 1]$, three actions, two shocks with probabilities $3/5$ and $2/5$, $\beta = 19/20$, stage reward $x - c_a$ with $c = (0, 3/20, 9/20)$, terminal payoff $x/2$, and revision weight $1 + x$. Its two affine successors for actions 0, 1, and 2 respectively are

$$(3x/4, 3x/4 + 1/20), \quad (17x/20 + 1/10, 17x/20 + 7/50), \quad (x/10 + 4/5, x/10 + 17/20).$$

TABLE VI
SAME-OBJECT COMPARISON ON THE PROSPECTIVELY FIXED ENVIRONMENTS

States	T	Method	Lower	Upper	Gap/upper (%)	Seconds
2	3	I	2.8414	2.8425	0.03	0.99
2	3	M	2.8414	2.8425	0.03	1.77
2	4	I	0.0000	0.0000	0.00	0.00
2	4	M	0.0000	0.0000	0.00	0.38
2	8	I	7.2534	9.0046	19.45	17.19
2	8	M	8.5290	9.0046	5.28	60.41
2	12	I	7.1419	7.5553	5.47	16.32
2	12	M	7.4446	7.5553	1.46	60.40
4	3	I	2.6157	2.6217	0.23	10.82
4	3	M	2.6206	2.6217	0.04	35.99
4	4	I	0.0233	0.0244	4.00	0.47
4	4	M	0.0235	0.0244	3.13	2.60
4	8	I	4.6631	7.8481	40.58	60.04
4	8	M	6.1990	7.8481	21.01	60.41
4	12	I	4.8936	6.6439	26.34	60.04
4	12	M	5.8606	6.6439	11.79	60.43

I: interval Bellman search. M: verified McCormick search. Bounds concern the identical common randomized Markov objective for each pair. Reported times include model construction and the operating solve; the search's 60-second cap is checked between iterations, not by interrupting a proof-producing node. A zero upper cost is an exact zero-cost case.

Full stop reasons, node counts, and memory appear in the supplement.

TABLE VII
COMMON RANDOMIZED MARKOV CERTIFICATES ON THE DEVELOPMENT COHORT

T	Cases	Zero-cost exact	Strict gain over K_D	Improved lower	Maximum gap/upper (%)
4	14	4	10	8	14.61
8	14	4	8	7	32.29
12	14	4	9	8	34.23

Strict gain means that a verified randomized upper cost is below the deterministic lower bound for the same initial law and constraints. It does not mean that the randomized optimum is attained by that candidate. Improved lower compares the connected local-support bound with the prior randomized support lower bound. All 42 uniform-initial rows and both nonuniform sensitivity tables are retained.

The seven installed rules, horizons 4, 8, and 12, and allowances 0.01, 0.05 are unchanged. Neural and spline rules are fixed computational incumbents, not estimated economic policies. Exact operating values are inexpensive in this model and their computation is charged. We do not use this cohort to claim performance when operating approximation itself is difficult.

The inherited deterministic comparison has 12 zero-cost and 12 positive-cost integrated equalities. Ten of the positive-cost equalities occur at horizon 4, two at horizon 8, and none at horizon 12. The maximum deterministic relative gaps at each horizon are reported alongside absolute gaps in the supplement. In the horizon-8 neural31003 case at $\varepsilon = 0.05$, the two initial cost functions are identical pointwise, but the arrays differ at later restart dates. Thus initial equality does not imply equality at every restart. The new diagnostic records the full rational disagreement locus.

The lotteries in (39) certify strict improvement over *every* feasible deterministic policy in 27 of the 42 uniform-initial configurations (Table VII). The horizon-12 defer case at allowance

TABLE VIII

OPEN RANDOMIZED INTERVALS AND GUARANTEED SAVINGS ARE DIFFERENT STATISTICS

T	Positive	Median (%)	Max. (%)	Strict gains	Min. gain	Max. gain
4	10	5.24	14.61	10	0.021898	1.041948
8	10	14.33	32.29	8	0.058701	0.896320
12	10	15.75	34.23	9	0.044193	0.747177

Width is $(U_R - L_R)/U_R$ among positive-cost intervals. Savings are $L_D - U_R$, not a difference between lower bounds. The last two columns condition on strictly positive savings. All 42 signed savings are in the supplement. These are exact re-tabulations of retained certificates, not new optimization runs.

TABLE IX

SELECTED NONLINEAR COMMON-MARKOV INTERVALS

T	Mesh per axis	Lower	Upper	Gap/upper (%)	Independent check
8	64	0.9438	3.9348	76.01	full
16	128	4.1908	7.6644	45.32	not independent
32	64	6.5389	16.7884	61.05	full
32	128	8.1264	12.5226	35.11	full R40
32	256	8.8718	11.8492	25.13	full R41

Uniform initial law on the square and operating allowance $1/2$. Inherited full independent scalar checks cover $(T, N) = (8, 64)$ and $(32, 64)$. R40 additionally replays the complete $(32, 128)$ object, including the candidate upper cost and all-restart operating bound. R41 subsequently independently rebuilt the complete $(32, 256)$ object; this revision incorporates that successful arithmetic audit with its source and proof hashes.

0.01, for example, has a randomized interval $[15.3172, 15.4670]$ and a positive deterministic separation. This addresses the economically relevant policy-class difference directly, without substituting a fixed-restart history-conditioned frontier. The 12 zero-cost cases are already optimal by feasibility and nonnegativity and are not counted as nontrivial search successes. All 30 positive-cost randomized intervals retain a reported nonzero width; the narrowest and widest relative intervals are distinguishable in the full table.

Changing the initial density to $2x$ yields 28 strict deterministic separations; changing it to $2(1 - x)$ yields 27. These are sensitivity results on the same developed environments, not independent observations. The connected local program improves 23 uniform lower endpoints, including the longer-horizon cases; it thus affects the primary global interval rather than a disconnected local diagnostic. It still does not establish global strong duality or close every interval.

9.7. Nonlinear two-state bounds

The retained nonlinear model has $(x, y) \in [0, 1]^2$, four actions, bilinear transitions, two shocks, reward $(x + y)/2 - c_a$, terminal payoff $(x + y)/4$, and revision weight $1 + (x + y)/2$. Its exact rational primitive specification is reproduced in the supplement. Whole-box operating witnesses and support supersolutions use outward arithmetic. The new lower construction uses 11 global support prices and five nonnegative local multipliers, including zero, at every date. It is paired with exactly the retained feasible upper policies. We run no new upper-portfolio competition and do not compare unequal proposal success fractions as evidence of optimality.

The old zero lower endpoint was not forced by the definition of revision cost: the operating constraints can force departures from the installed rule. Nonzero support supersolutions now

detect that restriction. At $T = 32, N = 256$, the interval is $[8.8718, 11.8492]$, with relative width 25.13%. This is a genuine improvement over $[0, 11.8492]$, but it is not a near-exact optimum or a high-dimensional scalability result. Both endpoints and the all-restart operating inequality are independently rebuilt at $T = 32, N = 64$, the R40 audit extends this to the complete $T = 32, N = 128$ object, and the R41 audit extends it to the finest $T = 32, N = 256$ object. The latter independently checks 92,274,688 support-action inequalities over 2,097,152 state–date cells; the original interval is unchanged. Memory for the retained support arrays grows as $8N^2(T + 1)(11 + 4)$ bytes before temporary arrays and interpreter overhead; the machine record reports the actual payload and process high-water mark.

The allowance $1/2$ belongs to the inherited exploratory nonlinear design. It is not made comparable to 0.01 simply by normalizing state variables: the operating scale and remaining horizon matter. We additionally compute lower-bound diagnostics at $\varepsilon = 0.05$ on four stated configurations. None is paired with a policy certified only at $1/2$. Without a newly feasible upper policy those tighter-tolerance rows remain lower-bound-only results. A failed sufficient box certificate is labeled uncertified, not infeasible. A box crossing an installed-rule threshold can charge the upper revision weight conservatively; it does not assert that every realized state in that box changes action.

10. VERIFICATION, ECONOMIC INTERPRETATION, AND LIMITS

A certificate has three distinct burdens. Its mathematical reduction must preserve the target policy class. Its finite arithmetic must check the asserted inequalities with the correct direction. Its publication must correspond to the checked objects. Hashes address the third burden, not the first two. Independent arithmetic addresses part of the second burden and can expose a model-contract error, but is not a proof that every economic assumption is appropriate.

The finite-state verifier uses SymPy rationals and imports neither search implementation nor the optimizer. It rebuilds the common Markov equations, primitive contracts, every policy-box split, every necessary contraction, the LP residual bounds, and the global leaf minimum. The continuous-state verifier independently checks the new lotteries, whole-cell costs, support supersolutions, and local primal vertices, using the prior independent primitive parser rather than the constructor’s piecewise-affine operations. The nonlinear verifier uses symbolic rational corner geometry and arbitrary-precision scalar integer induction instead of the constructor’s vectorized box operations. Its recorded coverage now includes the complete finest $(32, 256)$ object, not merely selected rows. The R42 structural checker separately reconstructs every new finite primal and dual certificate without importing its constructor. The verification report specifies precisely which objects passed; it never substitutes cross-run equality or a mutation test for an inequality proof.

The synthesis preserves all prior scientific material. Deterministic bounds, history-conditioned frontiers, the two-period analytic randomization example, adaptive local-LP analysis, 66 economic sensitivities, failed runs, and the stopped-control derivations remain available in full. The separate historical volume preserves the prior article and technical supplement verbatim; the current supplement supplies a direct map of the new results. The older 349-page historical annex remains unchanged as an archival object rather than a prerequisite for reading this article. The stopped consumption–portfolio problem is not relabeled solved: its inherited full-domain bound of approximately 7.1818 does not meet its 0.01 target. A killed Bellman comparison supports witness propagation but supplies neither missing action coverage nor a diffusion-discretization error bound (Kushner and Dupuis, 2001).

The economic conclusion established by the new primary calculation is precise. When randomized action assignment is admissible and costs attach to realized departures, common Markov

lotteries can strictly reduce the minimum implementation cost relative to every deterministic feasible rule. This conclusion holds on the certified configurations and initial laws. It does not estimate how frequently organizations permit such lotteries, how costly they are to administer, or the monetary welfare effect of a real deployment. Fixed randomization overhead, parameter uncertainty, and empirical calibration would require additional primitives and certificates. The joint-perturbation holdout study tests a fixed algorithm beyond its development seed, but is not a statistical generalization theorem.

10.1. Model uncertainty and the economic margin for randomization

The certificates can be translated into a robustness statement without treating designed numerical primitives as estimated data. Fix a nominal model, initial law, and policy class. Suppose a nonempty set of alternative models has verified uniform bounds

$$|J_t^{p,M} - J_t^{p,0}| \leq a, \quad |V_t^M - V_t^0| \leq a, \quad |C_0^{p,M} - C_0^{p,0}| \leq b \quad (45)$$

for every common policy p , state, and date, and every M in the set. Superscript 0 denotes the nominal model. The error recursion in Section 6 supplies such bounds from uniform reward, cost, terminal, and kernel enclosures on a common state space; no reset aggregation is needed for this fixed-space perturbation statement. Uniformity over policies is essential. A fitted residual checked only under the proposed policy does not establish (45).

PROPOSITION 4—Portable operating and cost intervals: *Let $\varepsilon > 2a$. A nominal policy feasible at allowance $\varepsilon - 2a$ is feasible at allowance ε in every alternative model. If L_+ is a nominal lower bound at allowance $\varepsilon + 2a$ and U_- is the nominal cost of such a tightened feasible policy, then every model-specific common-policy optimum belongs to*

$$[\max\{0, L_+ - b\}, U_- + b].$$

The same interval encloses the optimum that minimizes worst-case implementation cost over policies feasible in all the alternative models. In the latter assertion the upper endpoint uses a single policy, not a different policy for each model.

PROOF: For the proposed policy, $J^{p,M} \geq J^{p,0} - a \geq V^0 - \varepsilon + a \geq V^M - \varepsilon$. Conversely a policy feasible in a model M satisfies $J^{p,0} \geq J^{p,M} - a \geq V^M - \varepsilon - a \geq V^0 - \varepsilon - 2a$. Its nominal cost is at least L_+ and its cost in M at least $L_+ - b$. Take infima for the model-specific statement. For the robust problem, the first implication supplies one policy for all models and the second applies to every robustly feasible policy in each model. Taking the worst-case cost preserves the bounds. *Q.E.D.*

A strict randomized separation also has an economically interpretable administration-cost allowance. Let U_R be the certified cost of a feasible common lottery and L_D a lower bound for every feasible deterministic policy, for the same operating constraint and initial law. Suppose operating payoffs and dynamics are unchanged by administering the lottery. A nonnegative extra administration charge at date t bounded by h_t increases its expected discounted cost by at most $H = \sum_{t < T} \beta^t h_t$. If

$$U_R + H < L_D, \quad (46)$$

then the lottery remains cheaper than every deterministic feasible policy even after this additional charge. This conclusion does not require administration at every visited state; charging the

TABLE X
CERTIFIED ALLOWANCE FOR A PER-DATE RANDOMIZATION ADMINISTRATION CHARGE

T	Installed rule	ε	Det. lower	Lottery upper	h_* lower bound
4	defer	0.01	4.93668	4.73011	0.055684
4	defer	0.05	4.93206	3.89012	0.280857
4	neural31001	0.01	0.11117	0.08928	0.005902
8	defer	0.01	10.81072	10.63794	0.025667
8	defer	0.05	10.75071	9.85440	0.133151
8	neural31001	0.01	0.46864	0.40232	0.009853
12	defer	0.01	15.61281	15.46691	0.015872
12	defer	0.05	15.47129	14.72413	0.081278
12	neural31001	0.01	3.49393	3.36015	0.014553

Uniform initial law, unchanged development economy. Endpoints and the threshold are rounded outward. A positive threshold means that every smaller constant per-date overhead preserves a strict cost advantage over every feasible deterministic policy. All 126 rule/horizon/tolerance/initial-law margins are retained in the machine-readable results, including zeros.

bound everywhere is conservative. Deterministic rules are assumed not to pay the randomization-specific charge.

For a common per-date ceiling h , the certified break-even threshold is

$$h_* = (L_D - U_R)_+ \bigg/ \sum_{t=0}^{T-1} \beta^t. \quad (47)$$

Every $h < h_*$ preserves the strict ranking. Equality at h_* yields no strict conclusion. For model uncertainty, use a deterministic nominal lower bound at $\varepsilon + 2a$ and a nominal lottery feasible at $\varepsilon - 2a$; then replace the numerator by $(L_{D,+} - U_{R,-} - 2b)_+$. The current development results report $a = b = 0$: they calculate economic margins under known designed primitives, not empirical uncertainty radii.

11. CONCLUSION

A local operating requirement becomes a global revision certificate only when continuation compatibility and the cost of restoring feasibility are controlled together. Backward repair supplies that connection. A date-dependent allowance ladder covers finite undiscounted problems, while exact operating-action restrictions handle zero tolerance. These extensions retain the same common randomized Markov objective rather than replacing it by a necessary-action relaxation.

For endogenous finite-state dynamics, the operating-action disadvantage provides a probability bound, and implementation savings obey a regret cone. Their combination with reference-directed repair yields a quadratic small-allowance bound for a root relaxation at fixed action separation. This is a structural rate, not a polynomial guarantee for the unrestricted global problem. In the controlled audit, tighter relaxation geometry can materially improve a certificate, but difficult high-discount cases and solver stops remain in the evidence.

Observable-covariate disintegration supplies a separate controlled atomic continuum route. It preserves every Borel common policy by using a label recoverable from the current state, and its partition bounds converge when the finite fiber solves and primitive errors are refined. Exact exogenous aggregation remains a tractable complementary class; its twelve executed service economies retain tightly closed positive-cost intervals. The density exercise retains all primitive, solver, and repair terms, including a failed sufficient tightening condition.

These results do not make the original maintenance gaps disappear. Its 27 deterministic separations remain valid, all 30 positive-cost randomized intervals remain open, and the non-linear fine interval remains wide despite complete independent arithmetic verification. Generic completeness, a small-allowance rate, observable-fiber convergence, exact structural solvability, and a designed policy-class comparison answer different questions. Stating their assumptions and retaining their unsuccessful executions permits the method to advance without altering its economic objective or converting a certificate into an unsupported empirical claim.

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